12 · Solution Step 2: Keep the Hidden States

1. Goal of this step

Collect the right hidden states at the right moment, without adding extra computation.

2. What the paper does

While the LLM generates the search query token by token (greedy decoding, up to 32 tokens, using the KV cache), it keeps the **last-layer** hidden state of every generated token:

$$\mathbf{H} = [\mathbf{h}_1; \ldots; \mathbf{h}_n] \in \mathbb{R}^{n \times 4096}$$

Details from the paper:

Choice	Value
Which layer	Last layer only
Which tokens	Only the generated query tokens; special tokens (BOS, EOS, padding) are removed
Precision stored	float32
Max length	32 generated tokens (positions supported by the head: up to 128)

3. Why it is done this way

- **Last layer:** it has seen the most processing and sits right before word prediction, so it is the most "digested" view. (Other layers might work better, but the paper doesn't test this.)
- **Generated tokens only:** these are the tokens the LLM wrote *because* it wanted to search, so their states are focused on the search intent. Because the LLM is causal, each of them also looks back at the whole conversation.
- **Drop special tokens:** BOS/EOS/PAD states carry formatting information rather than meaning, and would pollute the average later (see `math/04_step_3_mean_pooling.md`).
- **Reuse during generation:** the model computes these vectors anyway. Keeping them costs memory but no extra forward pass.

4. Which problem it answers

Problem 2 (latency): no second model has to read anything.
Problem 3 (bottleneck): the states carry context that the text dropped.

5. What could go wrong

- **The first query token's state is "early":** it was computed before the rest of the query existed. The head's transformer layers (next file) are there to let later tokens inform earlier ones.
- **Storage during training:** 32 tokens $\times$ 4,096 numbers $\times$ 4

bytes ≈ 0.5 MB per trace. The paper caches 11,017 traces, which is manageable on disk but not free.

- **Tied to one LLM:** swap Qwen3-8B for another model (or even a fine-tuned Qwen3-8B) and the hidden states change, so the head must be retrained.

→ **Math:** the matrix $\mathbf{H}$ and its toy version are introduced in `math/02_step_1_input_projection.md`.